DERIVED PUSHFORWARD OF AN S-PERFECT OBJECT ALONG A FLAT PROPER FINITELY PRESENTED MORPHISM IS PERFECT AND COMMUTES WITH BASE CHANGE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- The formation of the derived pushforward of an S-perfect object commutes with arbitrary base change along cartesian squares. invoked at proof line 9198. Reduces the lemma to proving that Rf_*E is perfect.
- Noetherian approximation: schemes of finite presentation, their properness and flatness, and relatively perfect objects descend over filtered colimits of Noetherian rings after enlarging the index, by the Limits lemmas and the relatively perfect descent lemma. invoked at proof lines 9204-9213. Produces the Noetherian model X_i, E_i over R_i.
- Over a Noetherian base the derived pushforward of a relatively perfect object along a proper flat finitely presented morphism is perfect, by the perfect direct image lemma. invoked at proof lines 9214-9217. Supplies the Noetherian case the approximation reduces to.

Source reference. Original lemma 08IB: sources/stacks-perfect/pinned-original-source.tex, source lines 5362–5362. Cited at proof line 9198 to dispose of the base change half; the perfection half is inside the counted span.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cited at proof lines 9205-9207 to descend X to X_i; the descent chain is inside the counted span.

Source reference. Original lemma 081F: sources/stacks-chow/tags.txt, source lines 10215–10215. Cited at proof lines 9209-9210 to make X_i proper; the descent chain is inside the counted span.

Source reference. Original lemma 04AI: sources/stacks-chow/tags.txt, source lines 5617–5617. Cited at proof lines 9209-9210 to make X_i flat; the descent chain is inside the counted span.

Source reference. Original lemma 0DI8: sources/stacks-perfect/pinned-original-source.tex, source lines 9119–9119. Cited at proof lines 9211-9213 to descend E to an R_i-perfect E_i; the descent chain is inside the counted span.

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Source reference. Original lemma 08EV: sources/stacks-perfect/pinned-original-source.tex, source lines 6990–6990. Cited at proof lines 9214-9216 on the descended datum; the transfer back by base change is inside the counted span.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes which is flat, proper, and of finite presentation. Let $E \in D(\mathcal{O}_X)$ be S-perfect. Then Rf_*E is a perfect object of $D(\mathcal{O}_S)$ and its formation commutes with arbitrary base change.*

Proof. The statement on base change is Lemma 08IB. Thus it suffices to show that Rf_*E is a perfect object. We will reduce to the case where S is Noetherian affine by a limit argument.

The question is local on S, hence we may assume S is affine. Say $S = \mathrm{Spec}(R)$. We write $R = \mathrm{colim}\, R_i$ as a filtered colimit of Noetherian rings R_i. By Limits, Lemma 01ZM there exists an i and a scheme X_i of finite presentation over R_i whose base change to R is X. By Limits, Lemmas 081F and 04AI we may assume X_i is proper and flat over R_i. By Lemma 0DI8 we may assume there exists a R_i-perfect object E_i of $D(\mathcal{O}_{X_i})$ whose pullback to X is E. Applying Lemma 08EV to $X_i \to \mathrm{Spec}(R_i)$ and E_i and using the base change property already shown we obtain the result. $\square$